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SUPERSEDING
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MILITARY SPECIFICATION
CARTRIDGE CASE AND BASE PLUG
40MM, XM199E1

1. SCOPE

1.1 This specification is the applicable detail specification for Cartridge Case and Base Plug, 40MM, XM199E1.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposals form a part of this specification to the extent specified herein.

SPECIFICATIONS
Military
MIL-A-2550 - Ammunition and Special Weapons, General Specification for

STANDARDS
Federal
Federal Test Method Standard No. 151 - Metals; Test Methods

Military
MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109 - Quality Assurance Terms and Definitions
MIL-STD-1167 - Ammunition Data Card
MIL-STD-1169 - Packaging, Packing and Marking for Shipment of Inert Ammunition Components
MIL-STD-1235 - Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

DRAWINGS
Munitions Command
C10542395 - Case, Cartridge 40MM XM199E1
C10542393 - Plug, Base
IEL-10542395 - Index of Inspection Equipment Lists for Cartridge Case and Base Plug 40MM, XM199E1

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

FSC 1310

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2.2 Other publications.- The following document forms a part of the specification to the extent specified herein. Unless otherwise indicated, the issue in effect on the date of invitation for bids or request for proposal shall apply.

ASA-B46.1 - Surface texture

(Application for copies should be addressed to the American Standards Association 10 East 40th Street, New York, N.Y.)

3. REQUIREMENTS

3.1 General.- Materials, parts and assemblies shall comply with the requirements specified on the applicable drawings, referenced specifications, and the following:

* 3.2 Extrusion rod soundness.- The extrusion rods supplied for the manufacture of cartridge cases shall be free from cracks (including cracks existing in the ingot or billet and partially welded during extrusion) splits, inclusions and gross porosity.

3.3 Metal defects.- All components and assemblies shall be free from cracks, splits, cold shuts, inclusions, porosity or any other defects.

3.4 Mechanical properties.- The mechanical properties specified on the applicable drawings and referenced specifications shall be uniform throughout each component.

3.5 Protective coating.- Protective coating shall be in accordance with the requirements of the applicable drawing and specifications listed thereon.

3.6 Workmanship.- The requirements for workmanship are as specified on the applicable drawings, referenced specifications and the following:

3.6.1 Burr.- No part shall have a burr which might interfere with the assembly or function of the round, or which might be injurious to personnel handling the item.

3.6.2 Foreign matter.- No part or assembly shall contain dirt, chips, grease, oil, rust, corrosion or other foreign matter.

4. QUALITY ASSURANCE PROVISIONS

* 4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified

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in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Quality assurance terms and definitions.- Reference shall be made to MIL-STD-109 to define Quality Assurance terms used.

4.1.2. Inspection.- Inspection shall be in accordance with MIL-A-2550, except as specified herein.

4.2 First article sample.

* 4.2.1 Initial production sample.- At the beginning of regular production, a sample shall be submitted in accordance with contract requirements and shall consist of:

15 cartridge cases (coated)
15 cartridge cases (uncoated)
15 base plugs (coated)
15 base plugs (uncoated)

The sample shall be manufactured in the same manner using the same materials, equipment, processes and procedures as will be used in regular production. All parts and materials including packaging and packing shall be the same as used for regular production and shall be obtained from the same source of supply.

4.2.1.1 Examination and test.- After inspection and provisional acceptance at source, the sample shall be inspected for all the requirements of the drawings and specifications at a Government laboratory or such other facility specified in the contract.

4.2.1.2 Initial production sample failure.- Failure of the sample to comply with the requirements of the drawings and specifications shall result in sample disapproval.

4.3 Inspection provisions

4.3.1 Lot.

* 4.3.1.1 Lot Formation.- A lot shall consist of cartridge cases and base plugs produced by one manufacturer in one unchanged process, in accordance with the same drawings, same drawing revisions, same specifications and same specification revisions.

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4.3.1.2 Submission of product.- The product shall be submitted in accordance with MIL-STD-105 or MIL-STD-1235, as applicable.

* 4.3.2 Examination.- One hundred percent examination shall be performed for all critical defects. Examination for major and minor defects shall be performed on a defect basis in accordance with the classification of defects contained herein, using the sampling plans contained in MIL-STD-105. Continuous sampling plans in accordance with MIL-STD-1235 may be used if approved by the procuring activity. An AQL of 0.40 percent for each major defect and an AQL of 0.65 percent for each minor defect shall be used. All nonconforming material shall be rejected. Four power (4X) magnification may be used in performing visual examination. The roughness comparison specimens prescribed by ASA-B46.1 shall be used as a basis for surface roughness determination.

* 4.3.2.1 Tool control dimensions.- Dimensions marked tool control shall be gaged at the beginning of each eight hours production from each machine in operation and whenever tooling is replaced. Where destruction of components is necessary to inspect these dimensions, measurements of the tool may be substituted provided the contractor has established correlation between the tool dimensions and the component dimensions prior to the start of production. The contractor shall be responsible for maintaining adequate surveillance to assure that drawing requirements are being met throughout the life of the tool.

* 4.3.2.2 Case, Cartridge, Drawing C10542395

Defect	Method of Inspection
CRITICAL	
1. Propellant flash hole(s) missing or obstructed	Visual
MAJOR	
101. Diameter of sabot cavity not functionally concentric with outside body diameter	Gage
102. Diameter of flange not functionally concentric with outside body diameter	Gage
103. Diameter of base plug cavity counterbore not functionally concentric with outside body diameter	Gage
104. Diameter of base plug cavity not functionally concentric with base plug cavity counterbore	Gage
105. Rear face not perpendicular with outside body diameter	Gage
106. Diameter of propellant flash holes	Gage

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4.3.2.2 (Cont'd)

Defect	Method of Inspection
MAJOR (Cont'd)	
107. Diameter of body (outside)	Gage (see 6.3)
108. Diameter of sabot cavity	Gage
109. Diameter of flange	Gage
110. Diameter of base plug cavity counterbore	Gage
111. Diameter of base plug cavity	Gage
112. Depth from rear face to bottom of base plug cavity	Gage
113. Length from rear face to bottom of sabot cavity	Gage
114. Thickness of flange	Gage
115. Color improper	Visual
116. Metal defective	Visual
MINOR	
201. Depth to base plug cavity counterbore	Gage
202. Length, total	Gage
203. Location of propellant flash holes	Gage
204. Depth of undercut forward of flange (if applicable)	Visual (see 6.1)
205. Width of undercut forward of flange (if applicable)	Visual (see 6.1)
206. Radii or chamfers missing or incorrect	Visual (see 6.1)
207. Surface finish improper	Visual (see 6.1)
208. Protective coating incorrect, inadequate or damaged	Visual
209. Marking missing, incorrect or illegible	Visual
210. Burr	Visual
211. Foreign matter	Visual

4.3.2.3 Plug, Base, Drawing C10542393

CRITICAL	
1. Primer flash hole missing or obstructed	Visual
MAJOR	
101. Concentricity of forward outside diameter with flange	Gage
102. Concentricity of primer cavity with flange	Gage
103. Perpendicularity of rear face with flange	Gage
104. Diameter of propellant cavity	Gage
105. Diameter at front (outside)	Gage
106. Diameter of primer cavity	Gage

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4.3.2.3 (Cont'd)

Defect	Method of Inspection
MAJOR (Cont'd)	
107. Diameter of primer flash hole	Gage
108. Diameter of flange	Gage
109. Depth of primer cavity	Gage
110. Length from rear face to datum diameter	Gage
111. Length from rear face to front of flange	Gage
112. Thickness through base	Gage
113. Metal defective	Visual
MINOR	
201. Diameter of propellant cavity counterbore	Gage
202. Depth of propellant cavity counterbore	Gage
203. Length, total	Gage
204. Width of flange	Gage
205. Radii or chamfers missing or incorrect	Visual (see 6.1)
206. Surface finish improper	Visual (see 6.1)
207. Protective coating inadequate or damaged	Visual
208. Burr	Visual
209. Foreign matter	Visual

4.3.3 Maximum diameter gage.- Each cartridge case shall be gaged in a maximum diameter free fall chamber gage. Failure of the cartridge case to comply with the requirements of the applicable drawing shall be cause for rejection of the cartridge case.

4.3.4 Inspection testing.- The following tests shall be performed in accordance with 4.4.

* 4.3.4.1 Extrusion rod soundness.- Each rod extruded from each billet (ingot) shall be subjected to a Macro Etch Test to determine compliance with 3.2. Failure of any Macro Etched specimen to comply shall be cause for rejection of the rod represented by the specimen.

4.3.4.2 Mechanical properties.

4.3.4.2.1 Finished aluminum components.- Two cartridge cases from each lot of 10,000 or fraction thereof, after heat treatment, shall be subjected to this test. Failure of any cartridge case to comply with the mechanical properties requirements of the applicable drawing shall be cause for rejection of the lot.

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4.3.4.3 Protective coating.- A sample size of five base plugs and five cartridge cases from each lot shall be tested for compliance to the requirements of the applicable specification. Failure to comply with the requirements shall be cause for rejection of the lot represented by the sample.

4.3.5 Inspection equipment.- Index of Inspection Equipment List No. IEL-10542395 identifies the inspection equipment required to perform the examination and tests prescribed herein. Equipment design shall be in accordance with code designation on the applicable IEL. (Reference is made to Inspection Equipment List for definition of coding format.)

4.4 Test methods and procedures

* 4.4.1 Extrusion rod soundness.- A Macro Etched specimen shall be removed from each end of each rod after extrusion from the ingot or billet. The specimens, after suitable surface preparation, shall be macro etched in a ten percent solution of sodium hydroxide and water, heated to a temperature of 160 degrees F for a minimum of ten minutes, after which the specimens shall be visually examined to determine compliance with the requirements.

4.4.2 Mechanical properties.

4.4.2.1 Finished aluminum components.- From the samples selected in accordance with 4.3.4.2.1, after final heat treatment, one tension sest specimen conforming to the largest obtainable "F" type specimen as prescribed by Federal Test Method Standard No. 151, shall be taken in a circumferentially transverse direction forward of the web area. The criteria for performing the test shall be as prescribed by Federal Test Method Standard No. 151.

4.4.3 Protective coating.- The procedures for testing the cleaning of components and the application of protective coating shall be in accordance with the criteria for testing as specified in applicable specification.

5. PREPARATION FOR DELIVERY

5.1 Level C.- Minimum military pack

5.2 Packaging, packing and marking shall be in accordance with MIL-STD-1169.

5.3 Data card.- Data card information shall be as specified in MIL-STD-1167.

6. NOTES

6.1 Visual examination qualification.- When compliance with the applicable requirements is in doubt as a result of visual examination, this characteristic may be measured or gaged to determine acceptability.

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6.2 Ordering data.- Procurement documents should specify the title, number, and date of this specification.

6.3 If requirement cannot be met in free state the "go" inside diameter plug gage may be inserted in the cartridge case while inspecting the "go" and "not go" outside body diameter. If the go inside diameter plug is utilized the data cards shall be annotated accordingly.

* The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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